



Paper ID : 262100

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Subject Code:BBC207

Roll No:

BCA

(SEM. II) THEORY EXAMINATION 2025-26
ELEMENTARY MATHEMATICS BRIDGE COURSE

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	What is a prime number? Give two examples.	1	K3
b.	What is a singleton set?	2	K4
c.	Define Arithmetic Progression.	3	K3
d.	What is the order of a determinant? Find: $\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix}$	4	K3
e.	What is the order of a matrix?	4	K3
f.	State Pythagoras theorem.	5	K3
g.	Write the formula for sum of first n terms of a G.P.	3	K3

SECTION B

2. Attempt *any three* of the following:

07 x 3 = 21

Q no.	Question	CO	Level
a.	Explain prime and composite numbers. Find H.C.F. and L.C.M. of 12 and 18.	1	K3
b.	Show that the function $f(x) = 2x + 1$ is one-one.	2	K4
c.	Find the sum: $2+4+8+16+\dots+128$	3	K3
d.	If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ Find: $3A - 2B$	4	K3
e.	Prove that: $\sin^2 \theta + \cos^2 \theta = 1$	5	K3

SECTION C

3. Attempt *any one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain logarithms to base 10. Simplify: $\log(1000 \times 100) - \log 10$	1	K3
b.	State and explain the division algorithm with one example.	1	K3

4. Attempt *any one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Draw Venn diagrams for: $A \cup B$, $A \cap B$, $A - B$	2	K4
b.	Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ Check whether R is reflexive, symmetric and transitive.	2	K4

5. Attempt *any one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	State Binomial Theorem for positive integral index. Using Binomial Theorem Expand: $(x + 2)^5$	3	K3
b.	Define Harmonic Progression with examples. What is the relation between A.P. and H.P.?	3	K3



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6. Attempt any <i>one</i> part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Find transpose and inverse of: $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$	4	K3
b.	Evaluate: $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 3 & 1 \end{vmatrix}$	4	K3

7. Attempt any <i>one</i> part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Find area of triangle with base 8 cm and height 5 cm.	5	K3
b.	Define trigonometric ratios and explain with right triangle diagram.	5	K3